

Assignment: Code Clone Detection and Analysis

Objective

Select one of your own software projects, analyze it using a code clone detection tool, and propose improvements based on the findings.

Tasks

1. Choose one software project that you have personally developed.
2. Research and select a suitable code clone detection tool.
3. Run the selected tool on your project to identify duplicated or cloned code segments.
4. Analyze the detected code clones, including:
 - Their locations in the project
 - The type of clones detected (if supported by the tool)
 - Their potential impact on code quality and maintainability
5. Prepare a report that includes:
 - **Introduction** – Brief overview of the selected project
 - **Tool Overview** – Description of the chosen code clone detection tool and why it was selected
 - **Methodology** – Steps followed to analyze the project
 - **Results** – Clone detection findings, including screenshots or examples where appropriate
 - **Analysis** – Discussion of the detected code clones and their implications
 - **Proposed Solutions** – Recommendations for refactoring or other techniques to reduce or eliminate code duplication
 - **Conclusion** – Summary of findings and lessons learned

Deliverables

- A report (approximately **3–5 pages**) in PDF format.
- Screenshots or code snippets illustrating the detected code clones.
- A link to the analyzed project repository (if applicable).